

Statistics Learning Guide

Level 1 → Level 2 | Consolidated notes, intuition, calculations, tricks & A/B testing

Goal: understand the concepts in simple Hindi-English rather than memorizing formulas.

LEVEL 1 — STATISTICAL FOUNDATIONS

1. Standard Deviation (SD) & Z-score

SD tells how spread out observations are around the mean. **Z-score** tells how many SDs a value is from the mean: $z = (x - \text{mean}) / \text{SD}$.

Example: mean=50, SD=10, $x=70 \rightarrow z = (70-50)/10=2$. So 70 is 2 SD above the mean. Three SD below the mean = mean - 3×SD.

Trick: Z = “How far from the middle?” For an approximately normal distribution, about 68%, 95%, and 99.7% lie within ± 1 , ± 2 , and ± 3 SD respectively.

2. Skewness — Mean vs Median

Right-skewed: long right tail, typically mean > median. Left-skewed: long left tail, typically mean < median.

Trick: “Mean gets pulled toward the tail.” Example salaries ■20k, ■22k, ■25k, ■27k, ■2 lakh: the extreme salary pulls the mean upward, while median is more robust.

3. PMF, PDF, CDF

PMF: discrete exact probability $P(X=x)$. Example: 3 coin flips, $P(2 \text{ heads})=3/8$.

PDF: continuous probability density. A PDF value at one point is not itself a probability; probabilities come from area over an interval.

CDF: $F(x)=P(X \leq x)$. So CDF at 42 means probability of being 42 or less. For a continuous variable, probability at exactly one point is normally 0.

Trick: PMF=exact discrete probability; PDF=density/area; CDF=accumulated probability up to x.

4. Population, Sample & Sampling Distribution

Population = full group. Sample = subset. Example: 1,000,000 users are the population; 1,000 randomly selected users are a sample.

Sampling distribution = distribution of a statistic obtained by repeatedly taking samples and calculating that statistic.

Trick: One sample gives one estimate; repeated samples show how the estimate moves.

5. Central Limit Theorem (CLT)

Under common conditions, as sample size becomes sufficiently large, the sampling distribution of the sample mean becomes approximately normal even when the original population is not normal.

Real-world intuition: individual customer spending can be messy, but averages from many random samples tend to be much more predictable.

6. Standard Error (SE)

SD = spread of individual data. SE = uncertainty/spread of an estimator such as the sample mean. For a mean, $SE \approx s/\sqrt{n}$.

Example: $SD=30$, $n=100 \rightarrow SE=3$. If $n=900 \rightarrow SE=1$.

Trick: More data $\rightarrow$ smaller SE $\rightarrow$ less uncertainty.

7. Confidence Intervals

A confidence interval is an interval produced by a procedure designed to capture the true parameter at the stated long-run confidence level.

Example: sample average=80, margin of error=3 $\rightarrow$ interval 77 to 83.

Trick: estimate $\pm$ uncertainty. Avoid interpreting a particular 95% CI as a 95% probability statement about a fixed parameter.

LEVEL 2 — INFERENCE STATISTICS

1. Hypothesis Testing

H₀: baseline/no-effect claim. **H_a**: competing claim. Hinglish: **H₀** = “nothing changed”; **H_a** = “something changed.”

Example: **H₀**: $p_A=p_B$. **H_a**: $p_A \neq p_B$.

Reject **H₀** when evidence is sufficiently inconsistent with it at the chosen threshold; otherwise fail to reject **H₀**. Do not say “accept **H₀**.”

2. α and p-value

α is a pre-chosen Type I error threshold, commonly 0.05. The p-value is the probability, assuming **H₀** and the test model, of observing a result at least as extreme as the one obtained.

Rule: $p < \alpha \rightarrow$ reject **H₀**. $p \geq \alpha \rightarrow$ fail to reject **H₀**. **Trick:** p-value is not “the probability **H₀** is true.”

3. Type I, Type II & Power

Type I = reject a true **H₀** (false alarm). Type II = fail to reject **H₀** when a real effect exists (miss). Power = $1 - \beta$.

Trick: Type I = false alarm; Type II = missed signal.

4. One-tailed vs Two-tailed

One-tailed tests a pre-specified direction, e.g. $p_B > p_A$. Two-tailed tests for a difference either way, e.g. $p_A \neq p_B$.

Trick: one direction vs two directions.

5. t-test

t-tests are commonly used for inference about means when assumptions/model are appropriate. Variants include one-sample, independent two-sample, and paired t-tests.

One-sample form: $t = (\text{sample mean} - \text{hypothesized mean}) / SE$. For a one-sample t-test, $df = n - 1$.

Trick: t-test $\rightarrow$ think “compare means.”

6. ANOVA

ANOVA tests whether several group means are all equal versus the possibility that at least one differs. It compares between-group variation with within-group variation.

A significant ANOVA does not identify which pairs differ; follow-up post-hoc tests such as Tukey HSD can be used.

Trick: t-test commonly handles two means; ANOVA handles several means in one overall test.

7. Effect Size & Practical Significance

Statistical significance asks “is there evidence?” Effect size asks “how large?” Practical significance asks “does it matter?”

Example: 10%→12% = +2 percentage points absolute. Relative improvement=(12–10)/10=20%.

Trick: absolute = subtract; relative = change/original.

8. Power, MDE & Sample Size

MDE = smallest effect worth detecting. Example baseline 10%, important target 12% → MDE=2 percentage points.

- Smaller MDE → more users.
- Higher desired power → more users.
- Smaller α → generally more users.
- More users → smaller SE → easier detection.

Trick: Baseline=where you are; MDE=how much improvement matters; power=chance of detecting that specified effect.

9. Bootstrap / Resampling

Bootstrap repeatedly resamples the observed sample **with replacement**, usually keeping the same sample size, and recalculates the statistic.

Original [10,20,30,40,50]. Valid bootstrap sample: [20,20,50,10,40]. Another: [10,10,30,50,50].

Repeated bootstrap statistics form a bootstrap distribution used to estimate uncertainty, SE, and CIs.

Trick: sampling distribution = repeated samples from population; bootstrap = repeated resamples from your observed sample.

A/B TESTING — COMPLETE WORKED EXAMPLE

1. Business Question & Setup

E-commerce company tests a new checkout. A=Control/old checkout. B=Treatment/new checkout. Users are randomly assigned.

Goal: determine whether B truly improves purchase conversion rather than merely appearing better by chance.

Randomization trick: make A and B comparable on average so differences are more plausibly caused by treatment.

Group	Users	Purchases	Conversion
A — Control	100,000	10,000	10%
B — Treatment	100,000	12,000	12%

2. Conversion Rate & the Mean Trick

A: 10,000/100,000=0.10=10%. B: 12,000/100,000=0.12=12%.

Represent purchase as 1 and no purchase as 0. Then the mean equals the proportion. Example [1,0,1,0,1] → mean=3/5=0.60=60% conversion.

Trick: binary 0/1 mean = rate/proportion.

3. Observed Effect

Absolute improvement = $12\% - 10\% = +2$ percentage points.

Relative improvement = $(12 - 10)/10 = 20\%$.

Trap: +2 percentage points is not +2% relative.

4. H_0 and H_A

$H_0: p_A = p_B$ — “Dono versions actually same perform karte hain.” $H_A: p_A \neq p_B$ — “Dono different hain.” If the pre-specified business question is specifically whether B is better, $H_A: p_B > p_A$ is one-sided.

5. Pooled Proportion

For the two-proportion z-test under H_0 : $p_{\text{pool}} = (x_A + x_B)/(n_A + n_B)$.

$= (10,000 + 12,000)/(100,000 + 100,000) = 22,000/200,000 = 0.11$.

Why pool? H_0 assumes one common underlying conversion rate. **Trick:** H_0 says “same” → pool.

6. Standard Error

$SE = \sqrt{p_{\text{pool}}(1 - p_{\text{pool}})(1/n_A + 1/n_B)}$

$= \sqrt{0.11 \times 0.89 \times (1/100,000 + 1/100,000)} \approx 0.001399 \approx 0.14$ percentage points.

SE is the expected random noise in the estimated difference.

7. Z Statistic

$Z = (\text{observed difference} - \text{difference under } H_0)/SE$.

$= (0.12 - 0.10)/0.001399 \approx 14.3$.

Trick: $Z = \text{Signal/Noise}$. The 2-point difference is about 14.3 SEs from zero.

8. p-value & Decision

A two-sided $Z \approx 14.3$ gives an extremely small p-value, far below 0.05. Therefore **reject H_0** .

Correct interpretation: the data provide very strong evidence against equal conversion rates. Do not say the p-value is the probability H_0 is true.

9. 95% Confidence Interval

For the CI, use the observed-rate (unpooled) SE: $SECI = \sqrt{p_A(1 - p_A)/n_A + p_B(1 - p_B)/n_B} \approx 0.001398$.

$95\% \text{ CI} = (0.12 - 0.10) \pm 1.96 \times 0.001398 \approx 0.02 \pm 0.00274$.

$\approx [0.0173, 0.0227] = +1.73 \text{ to } +2.27$ percentage points.

The interval is entirely above zero, consistent with a positive effect.

10. Final A/B Interpretation

- Control conversion = 10%.
- Treatment conversion = 12%.
- Absolute effect = +2 percentage points.
- Relative effect = +20%.
- Pooled rate for test = 11%.
- $Z \approx 14.3$; p-value extremely small.

- 95% CI $\approx$ +1.73 to +2.27 percentage points.
- Statistical conclusion: reject H_0 .

Business conclusion still requires: effect size, CI, guardrails, cost, and practical significance.

A/B TESTING — SAMPLE SIZE & POWER

Planning Example

Baseline=10%. Important target=12%, so MDE=2 percentage points. Suppose $\alpha=0.05$, power=80%, and a two-sided two-proportion design. A commonly used planning calculation gives roughly **3,830 users per group** (about 7,660 total); exact values vary by method and design.

Do not memorize the long formula first. Understand the relationships: smaller MDE $\rightarrow$ much more data; higher power $\rightarrow$ more data; stricter $\alpha \rightarrow$ generally more data.

Analogy: shouting is easy to detect in a noisy room; a whisper needs more information. Large effect = loud signal; small effect = quiet signal.

A/B TESTING — EXPERIMENT DESIGN

Randomization Unit & Sticky Assignment

Decide whether assignment is by user, session, device, region, etc. User-level assignment is common. Sticky assignment keeps a user in the same variant to reduce contamination/carryover.

Sample Ratio Mismatch (SRM)

If planned split is 50/50 but observed split is 80/20, investigate before trusting the experiment. Possible causes: assignment bug, eligibility issue, filtering, or tracking problem.

Trick: planned split $\neq$ observed split $\rightarrow$ investigate.

Contamination, Novelty & Seasonality

Contamination: A users accidentally see B or vice versa, weakening the clean comparison.

Novelty effect: a new UI may create a temporary lift that fades.

Seasonality: behavior changes by day/week/holiday. Concurrent randomized experiments reduce this risk compared with comparing different time periods.

A/B TESTING — METRICS

Primary Metric

Main pre-specified metric. For checkout: purchase conversion rate. **Trick:** Primary = main reason for the experiment.

Secondary Metrics

Supporting metrics that help explain behavior: AOV, add-to-cart rate, checkout time, etc.

Guardrail Metrics

Metrics that should not materially worsen: refund rate, errors, latency, retention, complaints, revenue/user.

Trick: Guardrail = “don’t let something important get worse.”

Vanity Metrics & Multiple Testing

A metric can look impressive without business value. Also, checking many metrics and choosing whichever is significant increases false-positive risk. Define the primary metric before the test.

PRACTICAL SIGNIFICANCE

10.00%→10.01% can be statistically significant with a huge sample but economically unimportant. Conversely, 10%→15% can be practically huge but statistically inconclusive with too little data.

Four questions: Is it statistically significant? How large? How uncertain? Is it worth acting on?

END-TO-END A/B TESTING FRAMEWORK

Business question ↓ H_0 / H_A ↓ Primary metric + guardrails ↓ MDE + α + desired power ↓ Sample size ↓ Random assignment ↓ Run experiment ↓ Check SRM / contamination / tracking / seasonality ↓ Effect + SE + p-value + CI ↓ Practical & business significance ↓ Launch / Don't launch / Iterate

INTERVIEW MEMORY SHEET

- Z-score = how many SDs from the mean.
- Mean is pulled toward a skewed tail; median is more robust to outliers.
- PMF = exact discrete probability; PDF = continuous density; CDF = $P(X \leq x)$.
- CLT = repeated sample means become approximately normal under common conditions as n grows.
- SD = data spread; SE = estimator uncertainty.
- CI = estimate $\pm$ uncertainty.
- H_0 = baseline/no-effect; H_A = competing claim.
- p-value = probability of data at least this extreme assuming H_0 /model.
- Type I = false alarm; Type II = missed effect; Power = $1 - \beta$.
- One-tailed = one direction; two-tailed = either direction.
- t-test → means; ANOVA → several means.
- Effect size = how big; practical significance = whether it matters.
- Binary 0/1 mean = proportion.
- A/B: A=control, B=treatment.
- Randomization → comparable groups and stronger causal interpretation.
- MDE = smallest effect worth detecting.
- Smaller MDE / higher power / smaller α → generally more sample.
- Guardrail = protect against unintended harm.
- Never decide from p-value alone: combine evidence, effect size, CI, and business impact.

LEVEL 2 — REMAINING TOPICS

- A/B testing pitfalls: peeking, repeated testing, interference, Simpson's paradox.
- Multiple testing: FWER, Bonferroni, FDR, Benjamini–Hochberg.
- Chi-square tests and categorical data.
- Assumptions: independence, normality, equal variance, homoscedasticity.
- Non-parametric tests: Mann–Whitney U, Wilcoxon signed-rank, Kruskal–Wallis.

- Permutation tests and deeper resampling methods.
- Advanced experiment analysis.